

Integral Equation

$$x(s) + \int_{\Omega} k(s, t) f(t, x(t)) dt = y(s)$$



*Reduction to
Operator Form*

$$x + KN_f x = y$$

Operator Equations

$$\text{Linear: } [Kx](s) = \int_{\Omega} k(s, t) x(t) dt$$

$$\text{Nonlinear: } [N_f x](s) = f(s, x(s))$$

$$\text{Composed: } x + KN_f x = y$$

Solve using operator theoretic methods:

monotone operators, fixed points, Banach spaces